Services	Description
Cloud Coputing	
Capital Expense (CAPEX)	Money spent by a business to acquire, upgrade, or maintain **long-term assets** such as buildings, equipment, or infrastructure
Operational Expense (OPEX)	Ongoing costs for running day-to-day business operations, such as salaries, rent, utilities, or **cloud service usage**
Six Advantages of Cloud Computing	1. Trade CAPEX for OPEX 2. Benefit from massive economic scale 3. Stop guessing capacity 4. Increase speed and agility 5. Stop spending money on running and maintaining data centers 6. Go global in minutes
Global services & Regional services	1. **Global services = identity, security, DNS/CDN, account-wide management.** 2. **Regional services = compute, storage, databases, networking, analytics.**
IAM (Global Services)	
MFA device	**1. Virtual MFA App:** A software-based authenticator (e.g., Google Authenticator, Authy) on a mobile device generating time-based one-time passwords (TOTP). (Virtual device) **2. U2F Security Key:** A hardware key supporting the older FIDO U2F standard for strong two-factor authentication. (Physical device attached to system) **3. Hardware TOTP Token:** A physical device that displays time-based one-time passwords for AWS MFA. (Physical device not attached to system) 4. **SMS MFA (for root only):** Sends a one-time password via text message to a registered phone number (limited to the AWS account root user).
IAM Policies Structure	Consists of 1. Version**: policy language version, always include "2012-10 17"** 2. Id: an identifier for the policy (optional) 3. Statement: one or more individual statements (required) a. **Sid:** an identifier for the statement (optional) b. **Effect:** whether the statement allows or denies access (Allow, Deny) (mandatory) c. **Principal:** account/user/role to which this policy applied to (required in some circumtances) d. **Action:** list of actions this policy allows or denies (mandatory) e. **Resource:** list of resources to which the actions applied to (required in some circumtances) f. **Condition:** conditions for when this policy is in effect (optional)
EC2 Instances (Elastic Compute Cloud) (Regional Services)	
EC2 User Data	1. Bootstrap instances using an EC2 User data script 2. Bootstrapping means launching commands when a machine starts (only once) 3. Runs with the root user

EC2 Instance Types	1. General Purpose: For a diversity of workloads such as web servers or code repositories 2. Compute Optimized: compute-intensive tasks that require high performance processors 3. Memory Optimized: Fast performance for workloads that process large data sets in memory 4. Storage Optimized: storage-intensive tasks that require high, sequential read and write access to large data sets on local storage
Security Groups	1. All inbound traffic is blocked by default 2. All outbound traffic is authorised by default 3. Security groups only contain allow rules 4. Acts as a firewall at instance level
EC2 Instance Connect	1. A service that lets you securely connect to your **EC2 instances using temporary SSH keys generated** and managed through the AWS Management Console, CLI, or API, **without manually handling long-term key pairs.**
AWS Systems Manager Session Manager	1. A fully managed service that lets you securely **connect to and manage EC2 instances** or on-premises servers through an encrypted browser-based or CLI session **without needing SSH keys, bastion hosts, or inbound ports**.
Purchasing Options	1. On-Demand Instances – short workload, predictable pricing (Highest Cost) a. Linux or Windows - billing per second, after the first minute b. All other operating systems - billing per hour 2. Reserved (1 & 3 years) a. Reserved Instances – long workloads b. Convertible Reserved Instances – long workloads with flexible instances 3. Savings Plans (1 & 3 years) – commitment to an amount of usage, long workload 4. Spot Instances – short workloads, cheap, can lose instances (less reliable) 5. Dedicated Hosts – book an entire physical server, control instance placement. Purchasing Options a. On-demand – pay per second for active Dedicated Host b. Reserved 6. Dedicated Instances – no other customers will share your hardware 7. Capacity Reservations – reserve capacity in a specific AZ for any duration. No time commitment (create/cancel anytime), no billing discounts **Dedicated Host** = your **own physical server** with full hardware visibility and control. **Supports Bring your own license.** **Dedicated Instance** = isolated instance on dedicated hardware but **AWS manages the host**. **Don't support** Bring your own license**.**
EC2 Storage	
EBS Volume (Elastic Block Store) (Regional Services)	1. Network drive 2. locked to an Availability Zone 3. **Block based** storage

	4. Attach to a **single instance in the same AZ** 5. Encryption is not enabled by default
EBS snapshot	1. **incremental backup of an Amazon EBS volume**, stored in Amazon S3 2. So customer are billed only for the changed block store.
AMI (Amazon Machine Image, Regional Services)	1. AMI are built for a specific region 2. Can launch EC2 instances from a. A Public AMI b. Own AMI c. AWS Marketplace AMI 3. We must use AMI from the same region as EC2 instance. If AMI exists in different region we can copy that AMI to our desired region and launch EC2 instance.
EC2 Image Builder	1. Automatically create, build, test and distribute AMIs 2. Can be run on a schedule 3. Free Service
EC2 Instance Store	1. Temporary **block-leve**l storage **directly attached** to the host server for high-performance, **ephemeral** data that is lost when the instance stops or terminates. 2. Backups and Replication are customer responsibility
EFS (Elastic File System)	1. Managed NFS (network file system) that can be **mounted on 100s of EC2 instance across multiple AZs, region and VPCs** 2. **File** based storage 3. EBS can be used with windows EC2 but **not directly over native Windows SMB—you need the NFS client.** 4. Encryption is not enabled by default
EFS Infrequent Access (EFS-IA)	1. Storage class that is cost-optimized for files not accessed every day 2. EFS will automatically move files to EFS-IA based on the last time they were accessed
Amazon FSx	1. Launch 3rd party high-performance file systems on AWS 2. Fully Managed
FSx for Windows	1. Network File System for Windows servers 2. Supports SMB protocol & Windows NTFS
FSx for Lustre	1. High Performance Computing Linux file system
Elastic Load Balancing & Auto Scaling Groups	
High Availability	1. Run instances for the same application across multi AZ 2. Survive a data center loss (disaster)
Horizontal Scaling	1. Auto Scaling Group 2. Load Balancer
Scalability (vertical and horizontal)	1. Ability to accommodate a larger load by making the hardware stronger (scale up), or by adding nodes (scale out) 2. Vertical scaling (scale-up) involves adding resources to a single machine 3. Horizontal scaling (scale-out) involves adding more machines to distribute the load.
Elasticity	Once a system is scalable, elasticity means that there will be some “auto-scaling” so that the system can scale based on the load
Agility	New IT resources can be made available to developers within minutes

ELB (Elastic Load Balancer)	AWS managed load balancer. 4 types – 1. Application Load Balancer (HTTP / HTTPS only) – Layer 7 2. Network Load Balancer (ultra-high performance, allows for TCP) – Layer 4 3. Gateway Load Balancer – Layer 3 4. Classic Load Balancer (retired in 2023) – Layer 4 & 7
Scaling Strategies	1. Manual Scaling 2. Dynamic Scaling: Respond to changing demand 3. Predictive Scaling: Uses Machine Learning to predict future traffic ahead of time
Amazon S3(Regional Services)	
Buckets	1. Object based Storage 2. People store objects (files) in "buckets" (directories) 3. Globally unique name (across all regions all accounts) but buckets are created in a region
Objects	1. Objects (files) have a Key composed of prefix + object name a. s3://my-bucket/my_folder1/another_folder/my_file.txt
Security	1. User-Based (IAM Policy) 2. Resource-Based (Bucket Policies, Object ACL, Bucket ACL) 3. S3 buckets have encryption configured by default.
S3 - Versioning	1. Enabled at the bucket level 2. Protect against unintended deletes 3. Easy roll back to previous version
S3 – Replication	1. Must enable Versioning in source and destination buckets 2. Cross-Region Replication (CRR) - **compliance, lower latency access, replication across accounts** 3. Same-Region Replication (SRR) - **log aggregation, live replication between production and test accounts, abide by data sovereignty laws i.e. critical data must not leave country.** 4. Buckets configured for object replication can be **owned by same or different AWS account.**
Storage Classes	S3 automatically applies **server-side encryption (data at rest) with Amazon S3 managed keys (SSE-S3)** to every new object customer upload, **without any extra cost or configuration**. 1. Amazon S3 Standard - General Purpose 2. Amazon S3 Standard-Infrequent Access (IA) 3. Amazon S3 One Zone-Infrequent Access (One Zone - IA) 4. Amazon S3 Glacier Instant Retrieval 5. Amazon S3 Glacier Flexible Retrieval 6. Amazon S3 Glacier Deep Archive 7. Amazon S3 Intelligent Tiering Can move between classes manually or using S3 Lifecycle configurations
S3 Standard – General Purpose	1. Used for frequently accessed data 2. Low latency and high throughput **3. No data retravel charge** **4. No minimum storage duration charge**

	5. Use Cases: Big Data analytics, mobile & gaming applications, content distribution
S3 Standard - Infrequent Access	1. For data that is less frequently accessed, but requires rapid access when needed (i.e supports rapid data retrieval) 2. durable, **multi-AZ storage** for rarely accessed but important data. 3. Use cases: Disaster Recovery, backups
S3 One Zone-Infrequent Access	1. High durability in a single AZ; data lost when AZ is destroyed 2. Use Cases: Storing secondary backup copies of on-premise data, or data that can recreate
S3 Glacier Instant Retrieval	Millisecond access for rarely accessed data at the lowest retrieval latency but higher storage cost.
S3 Glacier Flexible Retrieval	Minutes-to-hours access with lower storage cost and optional bulk retrieval.
S3 Glacier Deep Archive	Lowest storage cost with retrieval times of 12–48 hours for long-term archiving.
S3 Intelligent-Tiering	1. Automatically moves objects between frequent and infrequent-access storage tiers to minimize S3 costs without impacting performance. 2. Store objects in two tires: a. One optimised for frequent access b. Another lower cost tier for infrequent access **3. No data retravel charge** **4. No minimum storage duration charge**
S3 Express One Zone	Ultra-low-latency, **single-Availability Zone S3 storage** class designed for frequently accessed data requiring microsecond retrieval.
AWS Snowball	1. Portable devices to collect and process data at the edge, and migrate data into and out of AWS 2. offline devices to perform data migrations
Edge Computing	1. Process data in an edge location i.e locations with limited or no access internet or computing power 2. Setup a snowball on edge location to preprocess data, machine learning, transcoding media.
AWS Snowball vs Edge Computing	1. **AWS Snowball** is a **physical data transfer device** for moving large datasets into/out of AWS 2. **Edge computing** processes and analyzes data **locally and deliver cloud services near the data source** to reduce latency and bandwidth use.
AWS Storage Gateway	1. Hybrid cloud storage service to **connect on premises environment with AWS Cloud**. 2. All data transferred between the gateway and AWS storage is encrypted. 3. Types: a. **Tape gateway** (Presents a virtual tape library for backup applications, storing virtual tapes in Amazon S3 or Glacier for cost-effective archival.) b. **Volume gateway** (Provides iSCSI block storage volumes that cache or back up on-premises data to AWS for disaster recovery.) c. **File gateway** (Offers a file-based interface (NFS/SMB) to store and access files as objects in Amazon S3.)
Databases Section	

Amazon RDS (OLTP, Online transaction Processing) (Regional Services)	1. Create databases in the cloud managed by AWS 2. Storage backed by EBS 3. Use SQL as a query language.
Amazon Aurora (OLTP) (Regional Services)	1. proprietary technology from AWS 2. PostgreSQL and MySQL are both supported. 3. Serverless DB
Amazon ElastiCache	1. managed Redis or Memcached
DynamoDB (Regional Services)	1. **key/value** (NoSQL) distributed "serverless" database with replication across 3 AZ 2. Single-digit millisecond latency – low latency retrieval
DynamoDB Accelerator - DAX	1. Fully Managed in-memory cache for DynamoDB 2. DAX is only used for and is integrated with DynamoDB, while ElastiCache can be used for other databases
DynamoDB – Global Tables	1. Make a DynamoDB table accessible with low latency in multiple-regions 2. Active-Active replication, cross region support (read/write to any AWS Region) 3. Automatically replicates DynamoDB tables across selected AWS Regions, enabling low-latency reads/writes and built-in disaster recovery.
Redshift (OLAP, online analytical processing)	1. Redshift is based on PostgreSQL 2. Columnar storage of data (instead of row based) 3. Pay as you go based on the instances provisioned 4. Used for analytics and data warehousing
Redshift Serverless	1. Automatically provisions and scales data warehouse underlying capacity 2. Use cases: Reporting, dashboarding applications, real-time analytics
Amazon EMR (Elastic MapReduce)	1. Create Hadoop clusters (Big Data) to analyze and process vast amount of data 2. The clusters can be made of EC2 instances with auto-scaling. Instances are managed by AWS.
Amazon Athena	1. **Serverless SQL** query service to analyze data stored in Amazon S3 2. Use cases: Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, CloudTrail trails, etc
Amazon QuickSight	1. Serverless machine learning-powered business intelligence service to create interactive dashboards from data stored in AWS services (like S3, RDS, Redshift) or external sources, and share insights securely at scale 2. per-session pricing
DocumentDB	1. MongoDB-compatible document database service 2. Fully Managed, highly available with replication across 3 AZ
Amazon Neptune	1. Fully managed graph database 2. Highly available with replications across 3 AZs
Amazon Timestream	1. Fully managed, serverless time series database
Amazon Managed Blockchain	Managed service to: 1. Join public blockchain networks 2. Or create your own scalable private network
AWS Glue	1. Serverless, managed extract, transform, and load (ETL) service 2. Prepare and load data for analytics

DMS – Database Migration Service	1. Migrate databases to AWS 2. The source database remains available during the migration 3. Supports both homogeneous and heterogeneous migration
RDS Deployment	1. Multi-AZ Deployment a. Main purpose is **high availability** b. Deployed in at least two AZs in a region c. The **standby** only takes over if the primary fails. d. **1 primary (writer)** instance handles **all write operations**. e. **Write/Read node – 1 (Same node used for both)** **Standby - 1** f. Non-Aurora: synchronous replication Aroura: asynchronous replication g. Endpoint of the DB remains same after failover. 2. Multi-Region Deployment a. Main purpose is **disaster recovery** and local performance b. Each region have multi-AZ deployment c. asynchronous replication **d. Writer Node – 1** **Reader Node – upto 15** 3. Read Replica a. Main purpose is **scalability** b. Can be within an AZ, cross-AZ, cross-region c. Write Node – 1 (use for both read/write) Read node – upto 15 No failover instance d. asynchronous replication
Other Compute Services	
ECS (Elastic Container Service)	1. Launch Docker containers on AWS 2. User provision & maintain the infrastructure (the EC2 instances) 3. AWS takes care of starting / stopping containers
Fargate	1. Launch Docker containers on AWS 2. Serverless offering i.e user do not provision the infrastructure (no EC2 instances to manage)
ECR (Elastic Container Registry)	1. Private Docker Registry on AWS 2. Store Docker images so they can be run by ECS or Fargate
EKS (Elastic Kubernetes Service)	1. Launch managed Kubernetes clusters on AWS
AWS Lambda	1. Event-Driven: functions get invoked by AWS when needed 2. Pricing: a. Pay per calls b. Pay per duration **3. Time limit**
Amazon API Gateway	1. A fully managed serverless service for creating, publishing, securing, and monitoring REST, HTTP, and WebSocket APIs at any scale. 2. Expose Lambda functions as HTTP API
AWS Batch	1. Fully managed batch processing at any scale 2. Batch jobs are defined as Docker images and run on ECS

	3. **No time limit** 4. Batch will dynamically launch EC2 instances or Spot Instances
Amazon Lightsail	1. Quickly deploying and managing virtual private servers, containers, databases, and networking with predictable, low-cost pricing. 2. **Has high availability but no auto-scaling**, limited AWS integrations 3. Cost effective monthly plan
Deploying and Managing Infrastructure	
CloudFormation (Regional Services)	1. Infrastructure as code 2. Model, provision, and manage AWS resources as code using templates for automated and repeatable infrastructure deployment. 3. Declarative programming (no need to figure out ordering and orchestration
AWS CDK (Cloud Development Kit)	1. An open-source framework that lets us define AWS infrastructure using familiar programming languages and then synthesize it into CloudFormation templates for automated deployment.
AWS SDK vs AWS CDK	**SDK** is for building applications that **use** AWS services, while **CDK** is for programmatically **creating and managing infrastructure** on AWS.
AWS Elastic Beanstalk (Regional Services)	1. A fully managed service that deploys and scales web applications automatically by handling infrastructure provisioning, load balancing, scaling, and monitoring. 2. Platform as a Service (PaaS) 3. Beanstalk is free but you pay for the underlying instances 4. Just the application code is the responsibility of the developer 5. Three architecture models: a. Single Instance deployment b. LB + ASG c. ASG only
AWS Elastic Beanstalk vs AWS Lightsail	1. **Amazon Lightsail** provides **simple, low-cost virtual private servers** with bundled compute, storage, and networking for straightforward apps or websites with no auto-scaling 2. **AWS Elastic Beanstalk** automatically **deploys and scales web applications** on managed EC2, handling provisioning, load balancing, and scaling for customer.
AWS CodeDeploy	1. Deploy application automatically (production) 2. Can deploy both in EC2 and on-premise
AWS CodeCommit (Deprecation)	1. AWS owned code repository, using the Git technology
AWS CodeBuild	1. Compiles source code, run tests, and produces packages that are ready to be deployed 2. Fully managed, serverless 3. Pay-as-you-go pricing – only pay for the build time
AWS CodePipeline	1. **Orchestrate** the different steps to have the code automatically pushed to production 2. Fully managed, compatible with CodeCommit, CodeBuild, CodeDeploy, Elastic Beanstalk, CloudFormation, GitHub, 3rd-party services (GitHub…) & custom plugins 3. Can deploy code in non-production env

AWS CodeArtifact	1. A fully managed artifact repository service for securely storing, publishing, and sharing software packages used in application development. 2. Developers and CodeBuild can then retrieve dependencies straight from CodeArtifact
AWS Systems Manager (SSM)	1. A unified operations service that helps you manage, automate, and gain visibility into your **AWS infrastructure and resources securely at scale**. 2. Need to install the SSM agent onto the systems we control 3. Send session log data to S3 or CloudWatch Logs 4. Provides automation features such as **Patch Manager**, which can automatically scan and apply OS patches to EC2 instances, reducing manual effort.
AWS Systems Manager Parameter Store	1. Secure, hierarchical storage service for configuration data and secrets (such as database strings, passwords, and license codes), enabling centralized management and retrieval with fine-grained access control.
AWS Systems Manager vs AWS Health Dashboard	1. **AWS Systems Manager** helps manage and automate infrastructure operations 2. **AWS Health Dashboard** provides real-time alerts and insights about AWS service events that may impact your account.
Global Infrastructure	
Regions	1. For deploying applications and infrastructure 2. Each region consists of a minimum **three** isolated and physically separated AZs.
Availability Zones	1. **One or more discrete data centers** with redundant power, networking and connectivity,
Edge Locations	Points of Presence for CDN
Amazon Route 53 (Global Service)	1. Route53 is a Managed DNS 2. The most common records a. A record (IPv4) b. AAAA IPv6 c. CNAME: hostname to hostname d. Alias
Route 53 Routing Policies	**Simple Routing** – Returns a single record for a domain. No Health Check **Weighted Routing** – Distributes traffic across multiple resources based on assigned weights. **Latency Routing** – Routes users to the region with the lowest latency. **Failover Routing** – Directs traffic to a primary resource, and automatically fails over to a secondary resource if the primary is unhealthy. **Geolocation Routing** – Routes based on the user's geographic location. **Geoproximity Routing (Traffic Flow only)** – Routes based on geographic proximity and optional bias settings. **Multi-Value Answer Routing** – Returns multiple healthy records to enable client-side load balancing.
Amazon CloudFront (Global Services)	1. Content Delivery Network (CDN) 2. DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall

CloudFront vs S3 Cross Region Replication	CloudFront: 1. Global Edge network 2. Files are cached for a TTL (maybe a day) 3. Great for static content that must be available everywhere S3 Cross Region Replication: 1. Must be setup for each region you want replication to happen 2. Files are updated in near real-time 3. Read only 4. Great for dynamic content that needs to be available at low-latency in few regions
S3 Transfer Acceleration	Speeds up uploads to S3 by routing data through Amazon CloudFront's globally distributed edge locations for faster transfer over long distances and then forward the data to the S3 bucket in the target region
AWS Global Accelerator	1. Improves the availability and performance of your global applications by routing user traffic through the AWS global network to the optimal AWS endpoint. 2. 2 Anycast IP are created for your application and traffic is sent through Edge Locations 3. Integrate with AWS Shield for DDoS protection 4. No caching, proxying packets at the edge to applications running in one or more AWS Regions 5. Good for non-HTTP uses cases such as gaming(UDP), IoT, VoIP as well as HTTP uses cases that requires static IP addresses.
AWS Outposts	1. It brings native AWS infrastructure, services, and tools to your on-premises or edge locations for a truly consistent hybrid cloud experience. 2. AWS Outposts are "server racks" that offers the same AWS infrastructure & services to build applications on -premises just as in the cloud 3. AWS will setup and manage "Outposts Racks" within our on-premises infrastructure
AWS Outposts vs Edge computing	**Edge computing** = the *approach* of running compute near the source of data. **AWS Outposts** = an *AWS product* that provides managed AWS infrastructure on-premises to enable that edge computing.
AWS WaveLength	1. delivers AWS compute and storage services to telecom 5G networks, enabling ultra-low-latency applications for mobile and edge devices. 2. Use cases: Smart Cities, ML-assisted diagnostics, Connected Vehicles, Interactive Live Video Streams, AR/VR, Real-time Gaming,
AWS Local Zones	1. Regional extensions of AWS that place compute, storage, and other services closer to large population centers for ultra-low-latency applications. 2. Extend VPC to more locations – "Extension of an AWS Region" 3. Good for latency-sensitive applications
Cloud Integration	
Amazon SQS (Simple Queue Service, FIFO)	1. Fully managed message queuing service that enables decoupled, scalable, and reliable communication between distributed application components. 2. Messages are deleted after they're read by consumers 3. Consumers share the work to read messages & scale horizontally

Amazon SNS	1. Fully managed **pub/sub** messaging service that sends notifications to multiple subscribers via **push** mechanisms such as email, SMS, or other AWS services. 2. Fully managed service for application-to-application (A2A) and application-to-person(A2P) communitcation.
Amazon Kinesis Data Streams	1. Managed service to collect, process, and analyze **real-time streaming data** at any scale 2. low-latency ingestion and processing of streaming data
Amazon Kinesis Data Firehose	1. Automatically loads streaming data into destinations like S3, Redshift, or Elasticsearch with minimal setup and near-real-time (buffered) delivery
Amazon MQ	1. SQS, SNS are "cloud-native" services: proprietary protocols from AWS 2. Managed message broker service for Apache ActiveMQ and RabbitMQ that enables applications to communicate using standard messaging protocols without managing the underlying infrastructure. 3. When migrating to the cloud, instead of re-engineering the application to use SQS and SNS, we can use Amazon MQ to work with existing system. 4. Amazon MQ has both queue feature (~SQS) and topic features (~SNS)
Cloud Monitoring	
CloudWatch Alarms	1. Alarms are used to trigger notifications for any metric 2. Use to create alarms to monitor estimate charge
CloudWatch Logs	CloudWatch Logs can collect log from a. Elastic Beanstalk b. ECS c. AWS Lambda d. CloudTrail based on filter e. CloudWatch log agents f. Route53 g. EC2: By default no logs, need to run a CloudWatch agent
Amazon EventBridge	1. **Serverless event bus** that lets us route real-time events from AWS services, SaaS applications, or custom sources to targets for event-driven application integration 2. Example: a. When an "OrderPlaced" event is published, EventBridge triggers a Lambda function to update inventory and send a confirmation email. b. On each S3 "ObjectCreated" event, EventBridge starts an AWS Glue job to process the new file.
EventBridge vs lambda function	EventBridge a. EventBridge is the **event bus** that captures and routes events. b. Think of it as a post office: it receives events (messages), decides where they should go, and delivers them. It does not process the messages itself. c. acts as the scheduler/trigger ("Hey, it's time to back up logs" or "A new log file arrived" lambda function a. Lambda is the compute engine that runs your code when an event arrives.

	b. Think of it as the worker who opens a message and does the job (runs code) whenever a message arrives. c. Lambda is the worker that actually performs the backup.
AWS CloudTrail	1. Records and monitors all API calls and account activity across your AWS infrastructure, providing audit logs for security, compliance, and operational analysis. 2. If a resource is deleted in AWS, investigate CloudTrail first
AWS X-Ray	1. Help developers **analyze and debug** distributed applications by tracing requests end-to-end to identify performance bottlenecks and errors.
Amazon CodeGuru	1. An ML-powered service for automated code reviews and application performance recommendations 2. Provides two functionalities a. CodeGuru Reviewer: automated code reviews for static code analysis b. CodeGuru Profiler: visibility/recommendations about application performance during runtime
AWS Health Dashboard	1. Provides real-time visibility into the operational status of AWS services and personalized alerts about events that might affect our AWS resources.
AWS Account Health Dashboard	1. Gives personalized view of AWS events and issues that specifically impact our account's resources, offering proactive alerts and remediation guidance.
VPC Section	
Elastic IP	1. Allows us to attach a fixed public IPv4 address to EC2 instance 2. all public IPv4 on AWS will be charged 3. IPv6 is free
VPC (Virtual Private Cloud)	1. A logically isolated **virtual network** in a single AWS Region 2. Two types: a. Public subnet - accessible from the internet b. Private subnet - not accessible from the internet 3. VPC is scoped to **a single AWS Region**, but it can span **all Availability Zones (AZs) within that Region**.
Subnet	1. A **subdivision of a VPC's IP range** that resides in **one specific AZ** 2. Subnets in a VPC can be in same AZ or in different AZ 3. Each subnet is scoped to **one specific AZ** and **cannot span multiple AZs**
Internet Gateway	Enables inbound and outbound traffic between VPC and the public internet for resources with public Ips i.e from public subnet
NAT Gateway	Allows private subnet resources to initiate outbound internet access (e.g., for updates) while preventing unsolicited inbound connections for protecting privacy.
NACL (Network ACL)	**1. Stateless**, subnet-level firewall in **AWS VPC** that controls inbound and outbound traffic using ordered **allow/deny rules**. **2.** Acts as a firewall at **subnet level.** **3. Rules** are evaluated in **ascending order (low to high) by rule number**; AWS checks each rule (allow or deny) until a match is found, and that decision is applied—if no rule matches, the default **DENY** takes effect.

Security Groups	**1. Stateful** virtual firewall at the EC2 instance level that controls inbound and outbound traffic to AWS resources using **allow rules**. **2.** Stateful means if EC2 instance sends a request, the response traffic for that request is allowed to flow in regardless of inbound security rule. **3.** Evaluate all rules before deciding whether to allow the traffic or not.
VPC Flow Logs	1. Capture detailed information about the IP traffic going to and from network interfaces in VPC 2. Helping you monitor, troubleshoot, and audit network connections and security.
VPC Peering	1. Connect **two VPC**, privately using AWS' network 2. Must not have overlapping CIDR (IP address range) 3. Not transitive
VPC Endpoints	1. Enable private, **secure connectivity between VPC and supported AWS services or on-premises resources without using the public internet**. 2. VPC Endpoint Gateway: S3 & DynamoDB 3. VPC Endpoint Interface: most services (including S3 & DynamoDB) 4. Secure **service-to-service traffic inside AWS**. (VPC to supported AWS Services)
AWS PrivateLink	1. provides private, secure connectivity **between VPCs and AWS services or third-party applications** over the AWS network, **eliminating the need for public IPs** or internet gateways. 2. PrivateLink is the technology; a VPC Interface Endpoint is the implementation of that technology.
Direct Connect (DX)	1. Establish a **physical connection between on-premises and AWS over a private network**. 2. Takes at least a month to establish
AWS VPN	1. lets us securely connect our on-premises network or devices to your AWS VPC enabling private communication between your data center and AWS resources. 2. AS VPN Comprises of two services: a. Site to Site VPN b. AWS Client VPN
Site to Site VPN	1. Connect an **on-premises VPN to AWS over the public internet** 2. **Site-to-Site VPN** creates an **IPsec tunnel between our on-premises network and an AWS VPC** 3. On-premises: must use a Customer Gateway (CGW) 4. AWS: must use a Virtual Private Gateway (VGW)
AWS Client VPN	1. fully managed, scalable, cloud based VPN service that enables secure, encrypted remote access for users to connect to AWS resources and on-premises networks from anywhere using OpenVPN-based clients. 2. Goes over public Internet 3. provides **secure, per-user remote access from individual devices to AWS or on-prem networks**. 4. Secures **user-to-VPC remote access**.

Transit Gateway	1. Fully managed hub service that simplifies network architecture by enabling us to connect **multiple Amazon VPCs and on-premises networks through a single central gateway**, reducing the complexity of many individual VPC peering connections.
Transit Gateway vs Direct Connect vs PrivateLink	1. **AWS Transit Gateway:** Connects multiple VPCs and on-premises networks through a **central hub** for scalable, managed routing. 2. **AWS Direct Connect:** Provides a dedicated, high-bandwidth **private network connection between your on-premises data center and AWS**. 3. **AWS PrivateLink:** Offers private access to specific AWS services or third-party applications over the AWS internal network without exposing traffic to the public internet.
Security & Compliance Section	
AWS Shield Standard	1. Free, managed service that automatically protects AWS applications from common network and transport layer DDoS attacks such as SYN/UDP Floods, Reflection attacks and other layer 3/layer 4 attacks 2. Activated automatically with no options for customisation so comes under AWS responsibility. 3. L3/L4 protection only.
AWS Shield Advanced	1. Premium DDoS protection service against more sophisticated attack — including detailed attack diagnostics, real-time monitoring, cost-protection for scaling during attacks, and 24/7 assistance from the AWS DDoS Response Team (DRT). 2. Customer need to subscribe for this and pay, so comes under customers' responsibility 3. L3/L4 protection + integration with WAF for L7 4. Protection for web application running in EC2, ELB, CloudFront, Route 53 and Global Accelerator.
AWS WAF (Web Application Firewall) (Global Services)	1. Managed security service that protects web applications and APIs from common web exploits—such as SQL injection, cross-site scripting (XSS), and malicious bots by allowing custom rules 2. Application-layer firewall specifically for web traffic. 3. Block or allow requests based on the **geographic location of the client's IP address** by using a **Geo match rule**
AWS Network Firewall	1. Fully managed, stateful network security service that enables us to deploy fine-grained firewall rules—including intrusion prevention, domain filtering, and traffic inspection—at the VPC level to protect entire Amazon VPCs from unauthorized access and network threats 2. Network-wide, stateful firewall for VPC traffic.
AWS Firewall Manager	1. Manage security rules in all accounts of an AWS Organization
AWS KMS (Key Management Service)	Fully managed service that allows us to create, manage, and control encryption keys supporting both AWS-managed and customer-managed keys
CloudHSM (Hardware security modules)	1. Fully managed service that helps us **safely create and use our own encryption keys** to protect sensitive data, while you keep full control over those keys 2. It gives us access to **dedicated hardware security modules** managed by AWS to manage and control encryption keys

Types of KMS Keys	1. Customer Managed Key: **create and fully control**, including permissions, rotation, and deletion by customer. 2. AWS Managed Key: created and controlled by AWS **on your behalf for a specific service** 3. AWS Owned Key: fully managed by AWS **for its internal use across many accounts** with no customer visibility or management 4. CloudHSM Keys
AWS Certificate Manager (ACM, Global Services)	1. Easily provision, manage, and deploy SSL/TLS Certificates 2. Supports both public and private TLS certificates
AWS Secrets Manager	1. securely stores and automatically rotates sensitive data like database passwords or API keys 2. Secrets are encrypted using KMS 3. **AWS KMS** creates and manages encryption keys used to encrypt or decrypt data but does not store the secrets themselves.
AWS Systems Manager Parameter Store vs AWS Secrets Manager	1. Parameter Store is for **general configuration and simple secrets**, while Secrets Manager is for **highly sensitive secrets with automatic rotation and auditing**.
AWS Artifact (Global Services)	Portal that provides customers with on-demand access to AWS compliance documentation and AWS agreements
Amazon GuardDuty (threat detection)	**Managed threat-detection service** that continuously monitors AWS accounts, workloads, logs and data for malicious activity or unauthorized behavior using machine learning and threat-intelligence feeds.
Amazon Inspector (vulnerability management)	**Scans resources only EC2, container images & Lambda functions** to identify software vulnerabilities and misconfigurations, **exposed network paths, and missing security patches**, helping you identify and remediate security risks.
AWS Config	1. A fully managed service that continuously tracks and records AWS resource configurations for auditing, compliance, and change management. 2. Helps record configurations and changes over time
AWS Macie	1. fully managed data security and data privacy service that uses machine learning and pattern matching to discover and protect your sensitive data in AWS 2. Identify and alert us about sensitive data, such as personally identifiable information (PII)
AWS Security Hub	A centralized service that continuously monitors, aggregates, and prioritizes security findings across all AWS accounts to help improve your security posture.
Amazon Detective	analyzes, investigates, and quickly identifies **the root cause of security issues** or suspicious activities (using ML and graphs)
AWS Abuse	Report suspected AWS resources used for abusive or illegal purposes. Abusive & prohibited behaviors are 1. Spam: receving undesired emails from AWS-owned IP address, websites & forums 2. Port scanning: sending packets to machine ports to discover the unsecured ones 3. DoS or DDoS attacks 4. Intrusion attempts: logging in on our resources

	5. Hosting objectionable or copyrighted content 6. Distributing malware: distributing softwares to harm computers or machines
Actions that can be performed only by the root user	1. Change account settings (account name, email address, root user password, root user access keys) 2. View certain tax invoices 3. Close your AWS account 4. Restore IAM user permissions 5. Change or cancel your AWS Support plan 6. Register as a seller in the Reserved Instance Marketplace 7. Configure an Amazon S3 bucket to enable MFA 8. Edit or delete an Amazon S3 bucket policy that includes an invalid VPC ID or VPC endpoint ID 9. Sign up for GovCloud
IAM Access Analyzer	A service that helps identify resources in our AWS environment that are shared with external entities, enabling you to analyze and refine access permissions.
Machine Learning Section	
Amazon Rekognition (Regional Scope)	A fully managed AI service that analyzes images and videos to detect objects, scenes, faces, text, and activities, and provides facial recognition and moderation capabilities.
Amazon Transcribe	1. Automatically convert speech to text 2. Uses a deep learning process called automatic speech recognition (ASR) 3. Supports Automatic Language Identification for multi-lingual audio
Amazon Polly	1. Turn text into lifelike speech using deep learning 2. Allow us to create applications that talk
Amazon Translate	Natural and accurate language translation
Amazon Lex	A service for building conversational interfaces using voice and text, powered by the same deep learning technologies as Alexa.
Amazon Connect	A cloud-based contact center service that enables businesses to provide customer support via voice and chat with scalability and ease of integration.
Amazon Comprehend	1. A fully managed NLP service that uses machine learning to **extract insights** such as sentiment, entities, key phrases, and language from text. 2. understands **what the text means**.
Amazon SageMaker AI	Fully managed service for developers / data scientists to build ML models
Amazon Kendra (document search service)	An AI-powered enterprise search service that enables organizations to **search across multiple data sources** with high accuracy using natural language queries.
Amazon Personalize	Fully managed ML-service to build apps with real-time personalized recommendations
Amazon Textract	1. Automatically extracts text, handwriting, and data from any scanned documents using AI and ML 2. extracts **text and data from documents**.
Account Management, Billing & Support	
AWS Organizations	1. Allows to manage multiple AWS accounts 2. Cost Benefits: a. Consolidated Billing across all accounts - single payment method

	b. Pricing benefits from aggregated usage (volume discount for EC2, S3…) c. Pooling of Reserved EC2 instances for optimal savings 3. Restrict account privileges using Service Control Policies (SCP)
Service Control Policies (SCP, Global Services)	1. Organization-level policies in AWS Organizations that define permissions guardrails, allowing or restricting AWS service actions across member accounts. 2. Does not apply to the Master Account 3. SCP is applied to all the Users and Roles of the Account, including Root user
AWS Control Tower	1. AWS Control Tower runs on top of AWS Organizations 2. A service that automates the setup and governance of a secure, multi-account AWS environment based on best practices. 3. automatically sets up AWS Organizations to organize accounts and implement SCPs
AWS Resource Access Manager (AWS RAM)	A service that enables you to securely share AWS resources across accounts or within an AWS Organization without duplicating them.
AWS Service Catalog	1. A service that lets organizations create, manage, and distribute approved collections of AWS resources (portfolios and products) to maintain consistency and compliance. 2. New AWS users may create non-compliant stacks, so service catalog is a self-service portal with admin-approved, pre-defined products ensures consistency and compliance.
Pricing Models in AWS	1. Pay as you go 2. Save when you reserve 3. Pay less by using more 4. Pay less as AWS grows
Advanced Identity Section	
AWS STS (Security Token Service)	Enables you to create temporary, limited- privileges credentials to access your AWS resources
Amazon Cognito	A service that provides **user authentication, authorization, and user management for web and mobile applications**, supporting sign-up, sign-in, and access control.
Microsoft Active Directory (AD)	A directory service that manages users, computers, and permissions in a Windows domain network, enabling centralized authentication, authorization, and policy enforcement.
AWS IAM Identity Center (Successor to AWS SSO)	1. A service that centrally manages **SSO access and permissions across multiple AWS accounts** and business applications. 2. one login for multiple AWS accounts & applications
AWS Compute Optimiser	A service that analyzes resource usage and provides recommendations to right-size **EC2 instances, Auto Scaling groups, EBS volumes, and Lambda** functions for optimal performance and cost efficiency.
Billing and Costing Tools (Global Services)	1. Estimating costs in the cloud a. Pricing Calculator 2. Tracking costs in the cloud a. Billing Dashboard (high-level snapshot)

	b. Cost Allocation Tags c. Cost and Usage Reports (raw, detailed data for custom reporting or advanced analytics) d. Cost Explorer (visual and interactive cost analysis) 3. Monitoring against costs plans a. Billing Alarms b. Budgets
AWS Pricing Calculator	1. A web-based tool to estimate and plan AWS service costs based on your expected usage. 2. **Estimate future costs** before **we deploy resources**.
Cost Allocation Tags	Use cost allocation tags to track your AWS costs on a detailed level
Cost and Usage Reports	1. contains the most comprehensive set of AWS cost and usage data available 2. lists AWS usage for each service category used by an account and its IAM users in hourly or daily line items, as well as any tags that we have activated for cost allocation purposes
Billing Dashboard	A central AWS console page that shows your current and forecasted costs, usage summaries, and payment details in one place
Cost Explorer	1. A tool to visualize, analyze, and track your AWS spending and usage **over time for better cost management**. 2. Enables us to predict future AWS costs by analyzing historical usage patterns 3. Must have to activate AWS generated tag and user defined tag to be appear on report.
Cost Explorer vs Compute Optimizer	**1. AWS Cost Explorer:** Analyzes and visualizes past and forecasted AWS spending. **2. AWS Compute Optimizer:** Recommends optimal EC2, EBS, and Lambda configurations to reduce cost and improve performance.
Billing Alarms	1. Billing data metric is stored in CloudWatch us-east1
AWS Budgets	1. Create budget and send alarms when costs exceeds the budget 2. 4 types of budgets: a. Usage b. Cost c. Reservation d. Savings Plans
CloudWatch vs AWS Budgets	1. **CloudWatch** = monitors **resource and application health**. 2. **AWS Budgets** = monitors **costs and usage against financial targets**.
AWS Cost Anomaly Detection	A service that uses machine learning to automatically detect unusual spending patterns and alert you to unexpected AWS cost changes.
AWS Service Quotas	1. A service that let us view and manage the resource limits (quotas) of AWS services, and request increases when needed. 2. Create CloudWatch Alarms on the Service Quotas console
AWS Trusted Advisor (Global Services)	1. A service that provides real-time guidance to help you **optimize costs, improve performance, enhance security, and increase fault tolerance** in your AWS environment. 2. Helps to identify low utilised EC2 instances

	3. 6 basic categories: a. Cost optimization b. Performance c. Security d. Fault tolerance e. Service limits f. Operational Excellence
AWS Support Plans	1. Basic Support Plan: free 2. Developer Plan 3. Business Support Plan 4. Enterprise On-Ramp 5. Enterprise
AWS Basic Support Plan	1. 24x7 access to customer service 2. AWS Trusted Advisor - Access to 7 core Trusted Advisor checks and guidance to provision resources following best practices to increase performance and improve security. 3. AWS Personal Health Dashboard - A personalized view of the health of AWS services, and alerts when your resources are impacted
AWS Developer Support Plan	1. All Basic Support Plan 2. Business hours email access to Cloud Support Associates along with general architectural guidance. 3. Unlimited cases / unlimited contacts
AWS Business Support Plan	1. Trusted Advisor – Full set of checks + API access 2. 24x7 phone, email, and chat access to Cloud Support Engineers 3. Access to Infrastructure Event Management for additional fee. 4. Provide technical support and general architectural guidance in context of **our specific use case** for production workloads. 5. Intended for **production workloads**
AWS Enterprise On-Ramp Support Plan	1. Intended for **production or business critical workloads** 2. Access to a **pool** of Technical Account Managers (TAM) 3. Concierge Support Team (for billing and account best practices) 4. Infrastructure Event Management, Well-Architected & Operations Reviews 5. Give detailed architectural guidance contextual to our application for critical workload (One per year) 6. Business-critical system down: < 30 minutes
AWS Enterprise Support Plan	1. Intended for **mission critical workloads** 2. Access to a **designated** Technical Account Manager (TAM) 3. Access to AWS Incident Detection and Response (for an additional fee) 4. Give most comprehensive and **contextual** architectural guidance contextual to our application. 5. Business-critical system down: < 15 minutes
AWS Architecting & Ecosystem	
Well Architected Framework 6 Pillars	1. Operational Excellence 2. Security 3. Reliability

	4. Performance Efficiency 5. Cost Optimization 6. Sustainability
Operational Excellence	Focus: **Efficient operations and continuous improvement** a. Automate changes, respond to events, and define standards b. Use infrastructure as code, runbooks, and post-incident reviews to improve processes.
Security	Focus: **Protecting data, systems, and assets** a. Enforce strong identity & access management (IAM) b. Apply data encryption in transit and at rest c. Enable logging and real-time monitoring for threat detection.
Reliability	Focus: **Workload recovery and fault tolerance** a. **Monitor and log** all system changes and performance. b. Design for failure with automation, multi-AZ deployments, and backups.
Performance Efficiency	Focus: **Optimal use of resources** a. Select the right compute/storage/network options. b. Continuously evaluate new AWS services and scale based on demand.
Cost Optimization	Focus: **Minimize cost and maximize value** a. Implement cost-aware architecture. b. Use right-sizing, auto scaling, and cost monitoring tools like AWS Cost Explorer.
Sustainability	Focus: **Reduce environmental impact** a. Optimize for energy efficiency. b. Choose the most efficient resources and regions.
Other Services	
Amazon Workspaces	1. fully managed, secure Desktop-as-a-Service (DaaS) solution that provides cloud-based virtual Windows or Linux desktops 2. great to eliminate management of on-premise VDI (Virtual Desktop Infrastructure)
Amazon AppStream 2.0	1. desktop application streaming service, fully managed. 2. delivers service to users via a web browser without any local installation.
AppStream 2.0 vs Amazon Kinesis	1. **AppStream 2.0** streams **applications** to end users. 2. **Amazon Kinesis** streams **data** for real-time processing and analysing.
AWS IoT Core	1. Easily connect IoT devices to AWS cloud. 2. Serverless
AWS AppSync	1. Store and Sync data across mobile and web apps in real-time using DB, Lambda etc for building backend 2. Use GraphQL (from facebook)
AWS Amplify	1. A set of tools and services that simplifies building, deploying, and hosting secure, scalable full-stack web and mobile applications on AWS.
Elastic Beanstalk vs Amplify vs Infrastructure Composer	1. Elastic Beanstalk is best for deploying and scaling **backend/server-based applications**. 2. Amplify is best for quickly **building and hosting front-end and serverless full-stack apps**. 3. AWS Application Composer is for architects who want to **design serverless architectures visually**.

AWS Infrastructure Composer	1. Visually design and build serverless applications quickly on AWS (drag and drop)
AWS Device Farm	1. Fully-managed service that tests web and mobile apps against desktop browsers, real mobile devices, and tablets 2. Have the ability to configure device settings according to our test scenarios. 3. Device will send reports, log, screenshots
AWS Backup	1. Fully-managed service to centrally manage and automate backups across AWS services 2. Cross-Region Backup 3. Cross-Account Backup (using AWS Organizations) 4. Pay for the backup storage we use and the amount of backup data we restore.
Disaster Recovery Strategies	The overall strategy and processes to restore IT systems and data after an outage 1. Backup and Restore (Cheapest) 2. Pilot Light (A minimal, continuously running **critical core services** in the recovery region that can be quickly scaled up to full production after a disaster. It requires customer to first deploy whole infrastructure and then scale out) 3. Warm Standby (A scaled-down but **fully functional copy of the production environment** that can be rapidly scaled to full capacity during failover. Can handle request immediately) 4. Multi-Site / Hot-Site (Costliest, Fully active production environments in multiple regions running simultaneously, enabling instant failover with no downtime.)
Elastic Disaster Recovery	1. A specific service that continuously replicates on-premises or cloud servers to AWS, enabling quick failover 2. Quickly and easily recover your physical, virtual, and cloud-based servers into AWS
AWS DataSync	1. Move large amount of data from on-premises to AWS 2. The replication tasks are incremental after the first full load
AWS DataSync vs snowball	1. **AWS DataSync:** Online service to transfer data between on-premises storage and AWS over the network. 2. **AWS Snowball:** Physical device for offline, large-scale data transfer to or from AWS when network transfer is impractical.
Cloud Migration Strategies	1. Retire: Turn off things you don't need 2. Retain: Do nothing for now 3. Relocate: Move apps from on-premises to its Cloud version. Move existing workloads to the cloud *as-is* at the infrastructure level (e.g., VMware to AWS) without changing apps or architecture. 4. Rehost "lift and shift": Simple migrations by re-hosting on AWS. No cloud optimizations 5. Replatform "lift and reshape": Not changing the core architecture, but leverage some Cloud optimization. (Ex: Instead of moving DB to EC2, use RDS) 6. Repurchase "drop and shop": Moving to a different product while moving to the Cloud

	7. Refactor / Re-architect: Reimagining how the application is architected using Cloud Native features **Relocate** = move the entire stack unchanged at the virtualization layer. **Rehost** = redeploy apps in the cloud with minimal tweaks. **Replatform** = migrate and make selective improvements for cloud efficiency.
AWS Application Discovery Service	A detailed discovery tool that automatically gathers server, application, and dependency data to plan and design complex migrations, including dependency mapping and migration wave planning. 1. Agentless Discovery 2. Agent-based Discovery
AWS Migration Hub	Central location to collect servers and applications inventory data for the assessment, planning, and tracking of migrations to AWS
AWS Application Migration Service (MGN)	The service that actually does the migration using replication agent.
AWS Migration Evaluator	A free assessment tool that collects high-level infrastructure and usage data to provide cost estimates and right-sizing recommendations for migrating workloads to AWS.
AWS Fault Injection Simulator (FIS)	1. A fully managed service for running fault injection experiments on AWS workloads 2. Based on Chaos Engineering, it helps to uncover hidden bugs and performance bottlenecks
AWS Step Functions	1. Build **serverless** visual workflow to orchestrate Lambda functions 2. Possibility of implementing human approval feature
AWS Ground Station	1. Fully managed service that lets you control satellite communications, process data, and scale satellite operations
Amazon Pinpoint	1. Scalable 2-way (outbound/inbound) marketing communications service 2. Supports email, SMS, push, voice, and in-app messaging 3. Ability to segment and personalize messages with the right content to customers 4. Amazon SNS or Amazon SES -- a. In SNS & SES you managed each message's audience, content, and delivery schedule In Amazon Pinpoint, you create message templates, delivery schedules, highly-targeted segments, and full campaigns
Amazon WorkLink	Secure mobile access to internal web apps, not on-prem compute.
APN Consulting Partner (AWS Partner Network)	A professional services firm in APN that helps customers **design, migrate, and manage workloads on AWS**, providing expertise in cloud strategy, architecture, and implementation.
APN Technology Partner	A company in the APN that provides **software solutions—such as applications, tools, or platforms**—that are hosted on or integrated with AWS to help customers build, manage, and optimize their cloud environments.
AWS Services support reservations to optimise cost	1. EC2 reserved instance 2. DynamoDB reserved capacity 3. ElastiCache Reserved Note

	4. RDS: by reserving EC2 instance 5. Redshift reserved node
AWS Connect	Contact center service designed to facilitate customer communication.
AWS Cloud Adoption Framework (CAF) foundational capability/perspective	1. **Business:** Aligns cloud strategy with business goals, value, and outcomes. 2. **People:** Prepares staff, roles, and culture for cloud transformation. 3. **Governance:** Establishes financial management, risk, and compliance controls. Application portfolio management. 5. **Platform:** Designs and builds the target cloud architecture and landing zone. 6. **Security:** Implements security, identity, and compliance best practices. 7. **Operations:** Defines processes for monitoring, management, and ongoing optimization. Performance and capacity management
AWS OpsWorks	1. configuration management service that uses **Chef or Puppet** to automate server provisioning, configuration, and application deployment across EC2 instances or on-premises servers.
CAF Transformation Journey	1. **Envision:** Define business outcomes and build a high-level cloud strategy. 2. **Align**: Align stakeholders, teams, and capabilities to the cloud vision and identify cross regional dependency 3. **Launch:** Execute pilot projects and initial migrations. 4. **Scale:** Expand and optimize cloud adoption across the organization.
AWS Acceptable Use Policy (AUP)	1. Amazon's official document that defines **what customers are allowed and not allowed to do** with AWS services.
AWS support center	**1.** **customer portal** where customers can **create and manage support cases**, access troubleshooting resources, view account or billing information, and communicate directly with AWS Support engineers. **2.** **direct help** from AWS Support staff.
AWS Knowledge Center	1. online library of **official troubleshooting articles, FAQs, and step-by-step guides** that help you quickly resolve common AWS issues and learn best practices. 2. **self-help library** of solutions.
AWS Serverless Services	1. Compute: a. Lambda b. Lambda@Edge c. Fargate 2. Storage a. S3 b. EFS 3. Data stores a. DynamoDB b. Aurora Serverless c. RDS Proxy 4. API Proxy a. API Gateway 5. Application Integration a. SNS b. SQS

	c. AppSync 6. Orchestration a. Step function 7. Analytics a. Kinesis b. Athena